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http://localhost:8888/notebooks/gen-ai/linear%20regression.ipynb

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JupyterLab Python 3 (ipykernel)

```
[143]: import pandas as pd
import numpy as np
np.random.seed(42)
house_size=np.array([500,1000,2200,2300,3500,4200,5000]).reshape((-1,1))
house_price=np.array([22000,32000,45000,52000,69000,76000,90000]).reshape((-1,1))
print(house_size)
print(house_price)
```

```
[[ 500]
 [1000]
 [2200]
 [2300]
 [3500]
 [4200]
 [5000]]
[[22000]
 [32000]
 [45000]
 [52000]
 [69000]
 [76000]
 [90000]]
```

```
[129]: data=pd.DataFrame({'size':house_size.flatten(),'price':house_price.flatten()})
print(data)
```

	size	price
0	500	22000
1	1000	32000
2	2200	45000
3	2300	52000
4	3500	69000
5	4200	76000
6	5000	90000

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[129]: data=pd.DataFrame({'size':house_size.flatten(),'price':house_price.flatten()})
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4	3500	69000
5	4200	76000
6	5000	90000

```
[130]: from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(house_size,house_price,test_size=0.3,random_state=42)#the next entry after train_test_split is input(hous
print("training set:",(len(x_train)))#print(f"training set:{len(x_train)}")
print("testing set:",(len(y_test)))#print(f"testing set:{len(x_test)}")
```

```
training set: 4
testing set: 3
```

```
[131]: from sklearn.linear_model import LinearRegression
model=LinearRegression()
model.fit(x_train,y_train)
#y=b0+b1.x
#b1
slope=model.coef_[0]
intercept=model.intercept_

print(f"price:{slope}x size +{intercept}")
print(slope)
print(y_intercept)
```

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```
model.fit(x_train,y_train)
#y=b0+b1.x
#b1
slope=model.coef_[0]
intercept=model.intercept_

print(f"price:(slope)x size +{intercept}")
print(slope)
print(y_intercept)

price:[15.22417154]x size +[14521.44249513]
[15.22417154]
[-41608.97435897]
```

[141]: #prediction

```
y_train=model.predict(x_train)
y_test=model.predict(x_test)
new_house_size=np.array([1500]).reshape(-1,1)
predicted_outcome=model.predict(new_house_size)[0] #[0] is used only to take the first value
print(predicted_outcome)

[37357.69980507]
```

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